

Tasks

1.

In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.

Ans:

Step 1: Open Cisco Packet Tracer and place two routers (Router A and Router B) and two PCs (PC-A and PC-B) onto the workspace.

Step 2: Connect PC-A to Router A, connect PC-B to Router B, and connect Router A directly to Router B using a serial or Ethernet WAN cable.

Step 3: Click on Router A, go to the CLI tab, and assign IP address 192.168.1.1 to its local interface and 203.0.113.1 to its WAN interface.

Step 4: Click on Router B, go to the CLI tab, and assign IP address 192.168.2.1 to its local interface and 203.0.113.2 to its WAN interface.

Step 5: Configure PC-A with IP 192.168.1.10 (Gateway: 192.168.1.1) and PC-B with IP 192.168.2.10 (Gateway: 192.168.2.1).

2.

Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.

Ans:

Step 1: On Router A, create a virtual tunnel interface by entering the command interface Tunnel0.

Step 2: Assign a private tunnel IP address to it by typing ip address 10.0.0.1 255.255.255.252.

Step 3: Bind the tunnel to the physical interface by entering tunnel source GigabitEthernet0/1 and pointing to the destination with tunnel destination 203.0.113.2.

Step 4: Repeat the process on Router B by creating interface Tunnel0 and assigning it ip address 10.0.0.2 255.255.255.252.

Step 5: Define the source and destination on Router B using tunnel source GigabitEthernet0/1 and tunnel destination 203.0.113.1.

Step 6: Add a static route on both routers (ip route) directing the opposite LAN traffic through the local tunnel IP address.

Verification: Open the command prompt on PC-A and type ping 192.168.2.10 to confirm that packets securely travel through the GRE tunnel to PC-B.

3.

Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.
Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.

Ans:

Step 1: Click on Router A, open the CLI, and create a user credential by typing username remoteuser password mysecretpass.

Step 2: Place a new individual PC on the workspace to act as a remote home laptop and connect it to the internet network.

Step 3: Click on this Remote PC, navigate to the **Desktop** tab, and open the built-in **VPN** client application.

Step 4: Fill in the configuration details by entering WorkVPN as the name and 203.0.113.1 (Router A's public IP) as the Server IP.

Step 5: Type remoteuser in the username field and mysecretpass in the password field, then click the **Connect** button.

The status display inside the application will change to "Connected," showing that the remote user has safely joined the branch network.

4.

List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

Ans:

Difference 1: Site-to-site VPN connects two entire stationary office buildings together, whereas Remote Access VPN connects an individual moving employee to the main office.

Difference 2: Site-to-site VPN uses dedicated hardware routers at both ends, while Remote Access VPN requires software client apps installed on user devices.

Difference 3: Site-to-site VPN remains permanently turned on, whereas Remote Access VPN requires the user to manually log in every time they want to connect.

Site-to-Site Advantage: It offers very fast data transfer speeds because it runs on high-end, dedicated network hardware routers.

Site-to-Site Disadvantage: It is completely inflexible and cannot be used by employees who are traveling or working from a cafe.

Remote Access Advantage: It allows employees to securely access office data from anywhere in the world using any internet connection.

Remote Access Disadvantage: It increases cybersecurity risks if a worker's personal laptop gets infected with malware or a password is leaked.

5.

Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

Ans:

Router A Commands:

```
RouterA(config)# interface GigabitEthernet0/0
RouterA(config-if)# ip address 192.168.1.1 255.255.255.0
RouterA(config-if)# no shutdown
RouterA(config-if)# exit
RouterA(config)# interface GigabitEthernet0/1
RouterA(config-if)# ip address 203.0.113.1 255.255.255.252
RouterA(config-if)# no shutdown
RouterA(config-if)# exit
RouterA(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
RouterA(config)# interface Tunnel0
RouterA(config-if)# ip address 10.0.0.1 255.255.255.252
RouterA(config-if)# tunnel source GigabitEthernet0/1
RouterA(config-if)# tunnel destination 203.0.113.2
RouterA(config-if)# exit
RouterA(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router B Commands:

```
RouterB(config)# interface GigabitEthernet0/0
RouterB(config-if)# ip address 192.168.2.1 255.255.255.0
RouterB(config-if)# no shutdown
RouterB(config-if)# exit
RouterB(config)# interface GigabitEthernet0/1
RouterB(config-if)# ip address 203.0.113.2 255.255.255.252
RouterB(config-if)# no shutdown
RouterB(config-if)# exit
```

```
RouterB(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
RouterB(config)# interface Tunnel0
RouterB(config-if)# ip address 10.0.0.2 255.255.255.252
RouterB(config-if)# tunnel source GigabitEthernet0/1
RouterB(config-if)# tunnel destination 203.0.113.1
RouterB(config-if)# exit
RouterB(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```